DeltaSieve: Targeted Visual Validation of Spreadsheet Transformations

Alexander Aue-Johr
Institute of Rural Studies
Thünen-Institute
Braunschweig, Germany
ORCID 0009-0001-8683-3630

***Abstract*—Native visual validation of spreadsheet transformations is expensive when every worksheet must be rendered. DeltaSieve selects potentially affected regions using differences in cell values, effective style properties, layout, and dependencies of conditional formatting and charts. Unknown dependencies retain explicit capture fallbacks. Paired native rebuilds and retained transformation controls distinguish comparison evidence from changes introduced by Excel itself. The system records selected coverage, exact pixel observations, process measurements, and reusable evidence. This working manuscript describes the method and its evaluation protocol, reporting completed bounded checks separately from ongoing corpus observations. Independent full-reference and mutation experiments remain necessary to quantify detection coverage and avoided rendering cost.**

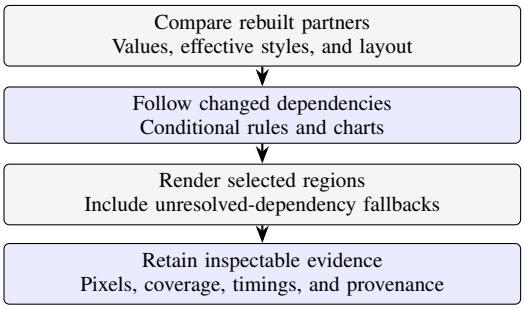


Fig. 1. DeltaSieve separates dependency-aware selection, native observation, and evidence. A proven unchanged dependency can exclude rendering. An unresolved dependency retains its recorded fallback. Targeted coverage remains explicit.

I. Introduction

A spreadsheet transformation can preserve cell values and still change the visible document. Removing an empty formatted cell can expose a background or alter automatic row height. Changing a referenced input can affect a distant conditional-formatting region or chart. Checking only edited cell payloads therefore leaves important transformation effects unobserved.

Native rendering supplies an operational way to observe these effects in the spreadsheet application that users employ. Exhaustive rendering, however, can be expensive for large sheets and corpora. Stored worksheet extents may also include empty or uniformly formatted tails. Dividing these extents by a fixed cell grid can create many images despite very little finite visible content. The challenge is to select useful rendering work while retaining explicit uncertainty about unsupported features and incomplete comparisons.

Figure 1 summarizes the selection and evidence workflow. DeltaSieve combines package-level change analysis with native visual comparison. It follows effective style identities across table renumbering, considers layout and dependency effects, and records the reasons for its capture choices. Supported conditional-formatting and chart dependencies can establish that an unchanged input cannot have changed the associated presentation under the supported model. Changed or unresolved dependencies remain capture candidates. The native stage measures geometry, renders bounded image batches, and compares retained evidence under a recorded state-pair policy.

The scientific question concerns the detection and cost of this selection. A dependency-aware planner is useful only if its reduced work retains the relevant effects in the evaluated domain. We therefore separate selector coverage, pixel observation and defect attribution. Independent full-reference samples and mutations test detection. Repeated same-input observations test rendering variability. A changed image remains an observation whose cause may require further evidence.

Existing spreadsheet dependency slicing motivates careful treatment of indirect effects [1]. Existing workbook comparison tools also provide reports of structural and cell-level changes [2]. DeltaSieve's evaluation must establish the additional contribution of its native-rendering selection and evidence workflow. An interactive Observatory and portable report make those observations inspectable. They support the method's evaluation without replacing its independent detection tests.

DeltaSieve grew out of the need to investigate visual regressions in SheetShrink, a workbook cleanup system described in a companion manuscript [3]. That manuscript studies cleanup rules and preservation structures. Here we study the selection, capture and evidence needed to evaluate rendered changes across workbook transformations, including alternative cleanup tools. The companion working PDF is included in the supplementary material at related-manuscripts/sheetshrink.pdf.

II. Preservation and Comparison Contract

A. Inputs, states, and observations

DeltaSieve validates a pair of spreadsheet states produced by a transformation. The package representation supplies evidence about where visible changes could occur. Native Excel rendering supplies the image observations for selected regions. The two sources of evidence serve different roles. A candidate is a reason to inspect a region; pixel inequality is an observation that still needs attribution.

The fast comparison uses Original and transformed workbooks after a full rebuild and save in their existing, separately owned Excel processes. This gives both partners the native recalculation step. Stored and ordinary-saved XML controls remain available for follow-up. Their changes explain potential causes without automatically enlarging the rebuilt-pair capture plan. An independently requested Check Performance comparison has its own transformed state and coverage.

B. Relevant visual area and candidate coverage

The relevant area includes values, formulas, effective visible formatting, text overflow, and objects. A union across relevant input states is needed to cover content removed by a transformation. Empty trailing geometry has a separate contract. The capture plan must retain enough evidence to distinguish a change inside the relevant area from a larger or smaller exterior canvas.

Candidate selection considers changed cell properties and layout, as well as regions affected through conditional formatting and graphical dependencies. Style and shared-string references require comparison of their effective contents because saving can renumber the corresponding tables. Supported reference reasoning can exclude a dependent region when its rule and inputs are unchanged. A changed dependency makes its affected region a candidate. Unresolved reference semantics require explicit fallback coverage.

A selected-region comparison does not by itself demonstrate that every relevant change in the workbook was selected. We evaluate that property using independently rendered reference areas and controlled mutations. The evaluation must record which reference cases completed and which exceeded their budgets. The paper's coverage claims are bounded by these observations and the supported dependency model.

C. Verdicts, failures, and reusable evidence

A successful comparison requires usable evidence from both partners. Missing hashes, failed native commands, and timeouts remain unsuccessful outcomes. Heartbeats establish add-in reachability and do not establish calculation or capture progress. Durable operation receipts and phase measurements support investigation of stalls.

The raw pixel and geometry observations remain attached to a finding after review. Product regressions, rendering variation, environment-dependent values, and verified capture-extent differences have separate classifications. A volatile formula path explains a possible cell-value change while other changed pixels in the same image can require further analysis. No global pixel tolerance is assumed.

Require: State pair (A, B), observation policy P
Ensure: Scoped comparison evidence and explicit outcomes
1: Record input identities and observation policy
2: Resolve value, effective-style and layout changes
3: Follow supported conditional and chart dependencies
4: Include conservative scopes for unresolved effects
5: Open owned native workers and prepare states under P
6: **for all** bounded packages of selected regions **do**
7: Measure native geometry in both states
8: Record zero-area outcomes and partition visible areas
9: Render paired tiles within pixel and memory budgets
10: Compare complete compatible image receipts
11: Retain differences, equality evidence and failures
12: **end for**
13: Record selected coverage and unfinished work
14: Release owned resources and record confirmed exits

Fig. 2. DeltaSieve's logical comparison stages. Missing or failed captures retain explicit outcomes. Independent reference controls evaluate selection.

Reusable observations bind the input states, relevant configuration, producing validation implementation, and declared rendering environment. Package-content identity is evaluated independently of ZIP entry order and compression. Reusing a completed study preserves its source manifest and compatibility checks. Portable reports retain the observations and their provenance for inspection without rerunning Excel.

III. Method

A. Observe an explicit state pair

Figure 2 summarizes the logical stages. Capture selection, native observation and outcome accounting remain separate operations.

The fast production strategy opens original and transformed workbooks in separate owned Excel processes and rebuilds them in place. A configured worker pool may lease existing processes across file pairs. Measurements must distinguish per-workbook opening from process and add-in initialization, and record the reuse policy when reporting campaign throughput. It compares their rebuilt states. Saved-input and pre-rebuild controls can explain changes introduced by Excel's own save or calculation behavior. Enabling additional controls incurs separate cost and must be recorded in the run configuration. A cached control is reusable only under its source and environment compatibility requirements.

Legacy XLS conversion precedes the XLSX comparison subject. A workbook requiring repair, inaccessible protected content or an unsuccessful save receives its own outcome. Missing evidence cannot yield equality. A host process existing in the operating system and an Office.js add-in responding are separate liveness signals. Neither proves that calculation or capture work has advanced.

B. Select potentially affected regions

The XML analysis compares cell values and formulas together with their effective style context. A style index is local to a package. The same index can identify changed properties, while different indices can identify equivalent properties. The

Resolve the style before selecting

Abbreviated XML control

```
<!-- Original package -->
<c r="B31" s="98"/>
<!-- Transformed package -->
<!-- B31 has no explicit cell record -->
<!-- Column B still supplies a fill -->
```

A removed empty cell can expose a column fill. Selection resolves the cell, row and column styles in both packages before deciding which region requires capture.

Style indices are package-local. Equal values or different indices alone do not establish equal formatting.

Fig. 3. An empty-cell deletion can expose inherited formatting. Abbreviated XML.

planner resolves that context before interpreting cell differences.

Figure 3 shows why a coordinate with no value can still require native observation. Inheritance determines the selected effect. The same reasoning applies when an explicit no-fill cell masks a colored column, or when a row style changes while the stored cell records remain identical.

Layout analysis includes row, column and default-format effects. Stored row-height attributes alone do not establish native effective heights. The native stage therefore measures the relevant geometry. Content bounds include values, formulas, visible fills and borders, and supported object extents. The union of both states is important when an operation removes content.

Consecutive changed inherited row formats are represented by one rectangle per contiguous run. Unchanged rows split these runs. Applying the same padding, clipping and capture grid preserves the tile union of individual row requests, while reducing intermediate candidate objects. This representation change does not establish a whole-workbook speedup or justify excluding a changed row.

Conditional-formatting analysis follows supported references. When a referenced input changes, its dependent application region becomes a candidate regardless of whether a particular rule outcome might remain the same. Unchanged supported rules and inputs can exclude a region. Unknown formulas, namespaces or extensions retain the documented fallback. Chart selection similarly considers supported formulas, referenced inputs and caches, with conservative handling of unresolved cases. The rule inventory and its unsupported cases are versioned with the tool.

C. Measure and render

Candidate regions are clipped to the justified content envelope. Connected regions are subdivided according to native measured image size. A fixed row count cannot predict the visible size of hidden or irregular rows. Zero-area regions have an explicit non-rendered outcome. Geometry and image batches have separate controls, allowing measurement overhead and rendering memory to be tuned independently.

Office.js image results become available after synchronization. Hash calculation therefore follows synchronization of the requested image batch. Compatible equal hashes can avoid transmitting and retaining equal images under the recorded transport policy. Actual unequal images remain evidence. Hash transport, image retention and scientific coverage are separate choices.

D. Keep exclusion and capture geometry consistent

The content envelope is a union over the inspected raw, saved and rebuilt states. Supported visible formatting contributes alongside values and formulas. An operation that removes content therefore cannot reduce the plan merely by making the transformed package smaller. Native object observations and unsupported dependencies retain their own coverage rules.

Height dependencies stop at the last row in this union. Policy v12 applies the same lower boundary to cell capture rectangles [4]. Context above the content and on either side remains available. Empty rows below the union cannot move content above them, whereas an empty row preceding later content can change its position. The boundary is recomputed through the existing state analysis. A later relevant extent enlarges it before capture selection. The planner reuses its stored rectangle and substitutes its unpadded bottom coordinate, adding no Excel operation or second worksheet scan.

This rule concerns the supported relevant-content envelope. It does not remove the need to model overflow, formatting or graphical objects correctly. Charts retain separate observations. A native zero-width or zero-height range has an explicit non-rendered outcome with geometry retained. A transition between visible and non-rendered states remains a difference. Agreement of two non-rendered ranges does not become a pixel-equality observation.

E. Two boundaries in the native capture protocol

The planner and native adapter enforce different boundaries. The Rust planner preserves the selected cell union while partitioning it under measured image budgets. The Office.js adapter must then return a complete image for each positive-area request. Both stages are required. A valid partition cannot make a silently truncated image complete, and a complete image cannot repair a selection that omitted a relevant coordinate.

Figure 4 shows the batch cardinality and all-state checks. Coalescing preserves holes and worksheet identity. The measured largest extent controls splitting, and every resulting child is checked again. Hidden spans can produce zero-area leaves rather than thousands of redundant image requests. Geometric observations remain in the evidence ledger.

Figure 5 shows the request/synchronization boundary. The implementation bounds both the number of queued images and their aggregate pixel cost. Queueing more work does not permit reading an unfinished result. After synchronization,

Every live state controls the split

Abbreviated Rust implementation

```
let measured = measure(&batch)?;
ensure!(
    measured.len() == batch.len(),
    "Incomplete geometry batch coverage"
);
for (candidate, geometries) in
    batch.into_iter().zip(measured)
{
    for geometry in geometries.values() {
        geometry.validate()?;
    }
    // Split if any live state is too large.
}
```

The planner measures the same candidate in every live state. A hidden range in one workbook can be visible in its partner. Any oversized state forces further partitioning.

A missing measurement is an error. Zero-area leaves retain an observation. A single oversized cell fails instead of disappearing.

Fig. 4. Every live state participates in geometry qualification. Abbreviated Rust.

Queue, synchronize, then consume

Abbreviated JavaScript implementation

```
const images = chunk.map(tile =>
  worksheet.getRange(
    CaptureGeometry.address(tile.raw)
  ).getImage()
));
await context.sync();
return images.map(image => image.value);
```

One synchronization completes the queued Office.js image requests. Returned images are then dimension-checked, cropped without resampling and finalized before another bounded batch.

The batch pixel limit also applies. A heartbeat while synchronization is pending does not prove that a capture completed.

Fig. 5. Office.js results are consumed after synchronization. Abbreviated JavaScript.

expected dimensions and crop geometry are checked before hashing or publication. A pending synchronization can therefore retain a live heartbeat without advancing verified capture progress.

Figure 6 records the visibility-transition control. No image is fabricated for its non-rendered state. A transition to positive area still produces a finding even though no paired pixels exist. This separates coverage, rendering presence and exact pixel

An absent image retains an outcome

Abbreviated visibility-transition receipt

```
{
  "geometry_differs": true,
  "rendering_differs": true,
  "pixels_compared": false
}
```

In the synthetic visibility-transition control, only the transformed state exposes a row. The original has a zero-area observation and no invented white image.

No paired pixels were compared. The rendering and geometry change remains a finding rather than an equality result.

Fig. 6. A visibility transition is retained without a fabricated paired pixel result.

comparison in the serialized result. The code panels abbreviate source contracts and omit orchestration mechanics. Their source revisions and hashes are retained in the supplementary evidence.

For captured images, raw dimensions and content origins remain available. Eligible exterior-size changes can be distinguished from changed overlapping pixels through the documented containment rules. A shifted origin, missing measurement or changed content pixel retains the applicable difference or incomplete outcome. Extent exclusion and exact white-margin exclusion have separate receipt flags. A global tolerance would erase that distinction.

F. Bounded execution and recovery

Resident workers receive commands over session-scoped WebSockets, while image uploads use bounded HTTP requests [5]. Durable commands, image receipts and results remain authoritative across reconnections. Duplicate delivery must preserve an in-progress command and acknowledge an identical completed result without running it again. Capture and upload overlap within bounded tile, pixel and encoded-memory windows. Office calls remain serial within an individual workbook.

Each comparison group owns its Excel processes and an isolated WebView profile. Cleanup records requested termination separately from confirmed exit. A retained clean-desktop control reproduced a peer taskpane restart under shared-profile cleanup; per-group isolation passed the corresponding two-group control and a four-group sequence. This observation motivates resource ownership in the tested environment. It does not establish that every startup delay is caused by stale profiles or registrations.

G. Separate control traffic from retained evidence

The resident command channel and the live status display have distinct roles. WebSocket delivery supports prompt command and completion notification, with durable records

supporting reconciliation after disconnection. The Observatory status stream starts with a snapshot and sends a restricted JSON Patch subset over allowed paths. Sequence gaps require resynchronization. Image uploads and detailed evidence remain HTTP operations [6].

A retained 100-card fixture records a 1,627,309-byte full status response, a 17,722-byte changed response and a 1,415-byte unchanged-comparison response. These measurements characterize status payloads for that fixture. They do not measure sustained comparison throughput. Bounded image queues and hash-based transport address separate costs and retain their own resource accounting. Neither a smaller status response nor a delivered heartbeat establishes that Excel has completed a pending render.

H. Retain outcomes and explain uncertainty

The report records selected regions, compared states, configuration, timings and failures. Pixel diagnostics provide changed-pixel counts, channel deltas and magnified regions. They do not assign a regression verdict on their own. Formula dependency traces can identify a supported path to a volatile function without explaining every changed pixel in the surrounding image.

Portable reports retain images using relative paths and integrity manifests. Their static viewer needs no Excel installation or running control API. Run metadata must be reviewed for local paths and publication-sensitive content before public release. Historical replay requires recorded event history and cannot reconstruct observations that were never retained.

New configurations retain equal captures by default. Packaging can replace a verified-equal PNG with a smaller lossless encoding after checking exact decoded RGBA equality. Differing or unverified image references preserve their original encoding. Live inspection uses a bounded window of capture receipts and the same spatial Atlas renderer. Capture glyphs take precedence over nearby heartbeat pulses in the worker timeline. These display events establish image availability, while the comparison report separately records the pixel verdict. Publication maps retained receipt identifiers to packaged images so that completing a job does not invalidate its recent inspection links.

Figure 7 shows recorded image availability in the same inspection surface as liveness. Opening a capture selects its recorded worksheet and range in Sheet Atlas. Image availability and a completed pixel comparison remain separately recorded events.

I. Collect independent runs in a static atlas

Each analysis can publish a static report with a compressed catalog, per-run detail and relative image assets [7]. A collection groups workbooks by declared corpus, technical name and original content identity. Processor versions and source runs keep separate identities. Missing input identity prevents automatic grouping. A merge checks retained evidence and preserves source provenance rather than collapsing independent observations into one synthetic run.

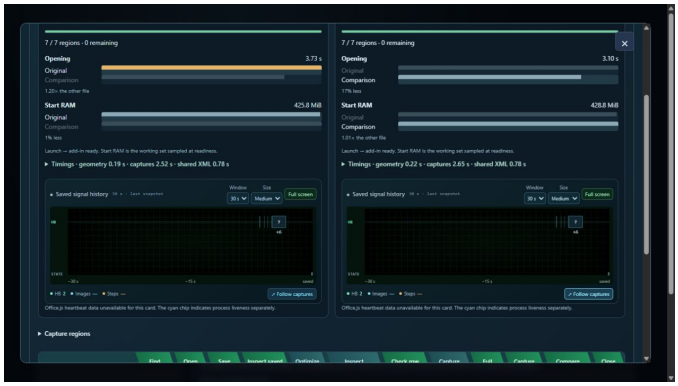


Fig. 7. Saved worker history from a completed seven-region native control. Capture glyphs share the heartbeat timeline and link to the corresponding retained images in Sheet Atlas. The displayed startup and memory measurements belong to this control and are not aggregate performance results.

The viewer retains the full worksheet inventory, including hidden and empty sheets, while loading detail and images on demand. Zoom changes the view over the same workbook and sheet hierarchy. Uncaptured regions have no reconstructed image. Different processors share an inspection context only when the retained original image, range, state, geometry and validation signature match. Native image pixels are inspected without rescaling. A changed capture policy can therefore leave an older processor without a matching context even when both runs concern the same workbook.

Compressed JSON is decoded in the browser, and the exported site needs static HTTP hosting. Integrity manifests retain report and image identities. Visual markings expose recorded differences and incomplete evidence; they do not replace the scientific coverage statement. The atlas supports inspection of the same frozen outputs that can subsequently be revalidated with the native tool. It introduces no additional detection claim.

J. Bound uniform backgrounds across column-width changes

A solid inherited fill spanning every worksheet column has no finite right edge. Repeated column definitions can partition that same fill solely because their widths differ. The planner resolves the existing column-format map and excludes those width-only boundaries from the background envelope when every column applies the same supported solid fill without borders. The existing byte scan supplies the column and style records. This guard adds neither another package read nor an Office.js synchronization.

Explicit cells, including blank overrides, remain in the envelope. Row-format transitions and finite content or merge extents remain bounded observations. Supported drawing bounds and conditional-formatting eligibility retain their independent checks. A differing fill, a border, an unknown style or unsupported worksheet context retains the applicable conservative extent. Column widths within the retained content still participate in layout comparison. The guard therefore

removes redundant background boundaries while preserving the observations used to detect interior changes.

An empty native `autoFilter` declaration is bounded separately. With a valid explicit cell rectangle, no filter children and only supported reference and revision-identifier attributes, its drop-down buttons remain within that rectangle. The planner includes this rectangle before applying the background proof. Actual filter criteria, sort children, unknown attributes and unsupported namespace contexts retain the conservative path. Native row geometry remains an independent observation. This distinction prevents a bounded button region from expanding an otherwise uniform background to every column.

The idle watchdog counts validated native row-height replies as geometry progress. The counter advances only after the returned height runs cover the requested spans and pass the numeric and ordering checks. Cached lookups and ordinary process or add-in heartbeats do not advance it. A workbook with many small sheets can therefore continue through a long measurement phase while a stalled native command remains bounded by its operation and idle deadlines.

Legacy drawing colour metadata is admitted only within a supported RGB colour node. The accepted extension declares the Office 2010 drawing namespace, marks that namespace as ignorable, and supplies a spreadsheet palette index between 0 and 65. The RGB value remains the drawing colour. Unknown namespaces, other attributes and out-of-range indices retain conservative coverage. The existing native frame, line width, transform and effect checks still determine whether the line admits bounded capture coverage.

Opening recovery retains explicit input provenance. Some converted Destatis workbooks contain both a legacy print-area or print-title name and the canonical built-in name in the same worksheet scope. Excel displays a name-conflict prompt. For a controlled recovery baseline, we rename the legacy alias only when exactly one canonical peer exists and package inspection finds no formula or relationship reference to that alias. Both definitions and their formulas remain present. Only the defined-name attribute in the workbook XML changes. Macro-bearing inputs and referenced aliases require separate review. Raw, normalized and transformed SHA-256 identities are retained with the changed member and alias mapping. Recovery results are labelled as comparisons against this normalized baseline.

Confirmed setup-phase completions also advance the idle-progress counter. Opening, saving and XML inspection may together exceed the idle interval while each phase completes successfully. A transition to a completed phase advances the counter once. Repeating the same completed status leaves it unchanged. Running statuses and liveness samples retain their existing watchdog semantics.

A supported horizontal line can have a different native vertical position from its serialized absolute transform after Excel resolves row anchors. Qualification requires a unique matching name, unrotated horizontal geometry, identical left coordinate and width, and a finite nonnegative native top coordinate. A single capture rectangle must contain the union of the native and stored positions, including half the stroke width and two points of rasterization context. A position outside that padded rectangle or a changed width retains the fallback. The resulting region is captured in both comparison states.

Capture policy version 13 reconstructs the visual-content envelope from the states actually compared after rebuilding. Native-qualified drawing bounds therefore replace earlier unqualified control bounds in this envelope. Opening/saving and input-transformation controls retain their XML change records and input identities. Their formatting extents do not enlarge the rebuilt-state image plan. The coverage claim is relative to the compared rebuilt states and the recorded selected regions.

Independent reference observations retain raw geometry and pixel outcomes. A derived extent classification compares exact RGBA pixels in the common rectangle. It admits an unpaired right or bottom extension only when every pixel exactly repeats the shared boundary column or row, respectively. A zero-area counterpart is classified as an empty extent only when every retained pixel on the other side is opaque white. This classification uses no pixel tolerance or image registration. The explicit plan and original observations remain available for review.

Each completed worksheet image comparison advances verified work progress after its pixel verdict is available. The comparison phase is marked active before that loop begins. This extends the same finite-work watchdog semantics through image decoding and pixel comparison. If an attempt fails, its remaining working evidence is moved alongside the failure logs. A record links this directory, including captures that were produced before a final comparison report existed. Incomplete evidence remains ineligible for completed-result reuse.

Capture policy version 14 also bounds repeating inherited backgrounds when conditional formatting has resolved finite application areas. The envelope includes every such area and a following context row before background-tail sampling is considered. Standard namespaces, resolved column information, supported worksheet features and the existing borderless solid-fill checks remain required. A rule that can affect the blank cells inside its application area is still captured there. Unresolved application references and application areas reaching the full worksheet boundary retain the conservative path.

In paired in-place execution, legacy coordinate projection is skipped when no pre-rebuild original image state exists. Each legacy comparison requires that state, so projecting its coordinates cannot produce an image observation. Serialized opening/saving controls and attribution records remain available. The rebuilt-state candidate plan and its native comparisons are independent of this omitted projection. This guard avoids work with no possible consumer; its end-to-end cost reduction requires a separate matched experiment.

Capture policy version 15 also recognizes unfilled column formats beneath supported solid row backgrounds. When every observed column style and the default style are known and borderless, and none has a visible fill, column width and font partitions add no horizontal edge to an empty row band.

TABLE I
FROZEN XLSX SOURCE INVENTORIES. SIZES ARE DECIMAL GB OF LISTED WORKBOOK FILES AND DO NOT MEASURE COMPLETED NATIVE COVERAGE.

Corpus	XLSX members	XLSX total (GB)
Data.gov	9,117	25.273
Destatis	1,182	2.364

The finite envelope, every explicit cell, row transitions and a following column remain selected. Visible column fills, borders and unknown styles retain their existing transition or conservative fallback handling.

IV. EVALUATION

RQ1 asks whether the selector captures the known visible effects in the tested mutation classes and independent reference sample. RQ2 measures how much native capture work and elapsed time it avoids. RQ3 examines the reproducibility and investigability of retained findings, including failures and rendering variation.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version. Legacy XLS conversion is reported separately from XLSX cleanup. A historical Enron development selection contains 20,872 inputs, with 20,503 prepared artifacts and 369 unsuccessful preparation outcomes at its recorded preparation checkpoint. Its selected visual scope contains 16,799 workbooks. This run ended after the worker-pool investigation. These counts describe a development selection and do not establish completed visual coverage. The replacement EDRM Enron v2 XML handover contains 34,065 distinct content-addressed candidates totaling 8.470 decimal GB of uncompressed payload. The imported selection unit is a distinct candidate SHA-256. Candidate transport was checked against the supplied manifests, and the import retained the original candidate bytes. This externally generated extraction includes legacy Excel and other OLE candidates. Native format recognition, successful XLSX conversion and visual eligibility remain separate evaluation stages. Candidate count therefore describes the imported input frame, with workbook outcomes to be reported after processing. The upstream extraction provenance and local import identities are retained alongside the experiment.

Microsoft Graph OLE objects require an additional eligibility distinction. Optional native PowerPoint conversion can recover their chart data into newly generated XLSX workbooks. Those workbooks have a different preparation history from successfully opened original Excel files and are reported as a separate derived-data population. Successful recovery or equality of cell values does not establish visual equivalence of the source Graph chart or coverage of its parent presentation. They do not contribute visual-equality observations without a separately executed native comparison.

The two frozen XLSX inventories in Table I come from the versioned source profiles. Each listed archive member has an individual SHA-256 identity. Their source-frame sizes are separate from the eligible, completed and independently referenced subsets.

A dated evaluation cutoff will report compared workbooks, differences, failures, unsupported representations and unfinished work. Repeated attempts have their own counts. An unchanged package can bypass visual work under an exact input-identity check, but a failed capture cannot contribute an equality observation. The full recursive Enron XML-derived corpus is a separately identified population.

B. Sampling and reference controls

The random evaluation sample is selected from a frozen eligible frame with a recorded seed before inspecting visual findings. Strata use corpus identity, package size and structural features available before validation. Low-reduction and unchanged files remain represented. A separately labelled development stress set contains large and historically problematic workbooks.

Feasible sampled files receive full native reference captures over the relevant visual area using a capture plan independent of the candidate selector. A full-reference timeout remains in the denominator. It is not replaced silently by an easier workbook. Mutations with known visible effects cover cell values, styles, geometry, conditional dependencies and objects. Mutations that Excel normalizes away are reported separately.

A reconciled historical Enron development sample retained twelve seeded source occurrences. Seven completed native references over 33 declared rectangles, two failed image-size preflight and three required unsupported graphical or embedded-object envelopes. All 33 completed rectangles had identical overlapping RGBA pixels. Two had additional opaque-white exterior strips, totaling 196,657 pixels, which were retained as raw extent differences. This sample is not held out, and failed or unsupported references remain in its denominator.

A separate twelve-occurrence Destatis sample used the same fixed-seed, size-quartile selection policy. All twelve selected packages contain graphical or embedded parts and remain unsupported by the current dimension-only reference preparer. They were not replaced by easier inputs. An optional native-object qualification extension admitted one of the same twelve inputs over two fixed areas. Its image rectangle fit inside the cover-sheet area, and both reference areas had identical overlapping RGBA pixels. The data-sheet image had a 20-pixel-high opaque-white exterior strip totaling 25,820 pixels. The remaining eleven inputs retained unsupported outcomes for legacy drawings, embedded objects or drawing effects. This extension checks native object rectangles, with explicit rejection of unsupported types and out-of-area objects. It does not certify decorative ink beyond those rectangles or establish corpus recall.

C. Detection and cost metrics

A completed independent reference determines whether a mutation has a visible effect under the recorded observation

policy. A selector miss occurs when such an effect falls outside selected coverage. Detection is measured per mutated workbook pair and per mutation class. A partially covered effect retains a separate outcome even when another difference makes the pair detectable. Pixel coverage requires a documented mapping between reference and selected capture coordinates, including extent differences.

Selection overhead counts candidates whose completed paired images are equal. This measures avoidable work within the tested policy. Pixel findings retain their attribution as transformation effects, rendering variation or unresolved differences. Failed references, missing partner images and timeouts retain explicit outcomes. Report both completed-reference detection and the fraction of all selected cases with usable reference evidence.

D. Mutation strata and statistical scope

The mutation matrix covers direct values and formulas, effective styles, conditional dependencies, chart inputs and objects, row/column geometry and visibility, and unresolved references. Serialization-only controls exercise equivalent package representations. Supported cases and fallback cases have separate expectations. Workbook identities, generator seeds, intended edits, reference areas and observed native effects identify every case. Spreadsheet errors such as division by zero are valid calculation outcomes and need their own fixtures alongside malformed or unopenable packages.

Repeated identical-input native controls characterize renderer variation. Volatile-function cases include NOW and RAND, with INDIRECT and OFFSET also requiring reference-policy controls [8]. The full rebuild and save policy applies to both states. An unchanged formula can therefore produce different rebuilt values. Such observations retain their raw pixels and calculation context for attribution.

Report sample counts and binomial uncertainty for independently sampled units within each declared population [9]. For zero misses in n independent trials with common miss probability, the exact one-sided 95% upper bound is $1 - 0.05^{1/n}$. This is approximately 9.5% for 30 trials and 5.8% for 50. Multiple mutations of one workbook share a cluster and do not establish that many independent workbook observations. Development stress cases remain separately labelled. Population estimates must account for the sampling strata and inclusion probabilities.

E. Matched cost accounting

Cost measurements separate XML planning, geometry, native rendering, transfer, image comparison and persistence where instrumentation permits. Report image count, rendered pixels in each state, compared pixels and retained bytes alongside elapsed time. Overlapping captures count as repeated rendering work; coverage-union size is reported separately. Matched selective and independent reference runs use the same state-preparation policy, environment, image budget and reuse configuration. Analysis, opening, rebuild and save remain part of the declared end-to-end timing even when capture-phase timings are also reported. Batch and concurrency experiments use matched inputs and repeated runs. A changing production queue does not identify the causal benefit of a tuning parameter. Matched timing summaries report paired distributions and repeated-run variation. Full-reference timeouts remain censored observations alongside completion counts. Full-corpus cost estimates use measured visible geometry and stratified samples, with uncertainty and assumptions about hidden regions and object extents.

F. Bounded opening experiment

A retained development experiment opened four copies of a synthetic workbook with eight sheets, each containing one million rows and three numeric columns. The same Windows 11 host used a Ryzen 7 7800X3D. Two reversed-order rounds compared sequential and simultaneous independent Excel processes on a shared private desktop [10]. Median batch times were 118.388 and 41.714 seconds, respectively. The measured endpoint includes verified Office.js add-in readiness. Staging, teardown and launch barriers are excluded, while caches and concurrent background work remain potential influences. This bounded observation supports overlapping startup; it does not measure arbitrary COM-command parallelism or full-corpus comparison throughput.

G. Geometry and capture batching

A warmed-up development experiment measured 64 ranges in one isolated Excel instance [11]. The two scalar passes each took 16.094 seconds. The two intervening batch passes each took 0.250 seconds. All returned geometries matched. A subsequent capture sequence used twelve regions, including four zero-area regions. Requested package sizes of 1, 2, 3, 4 and 8 took 3.125, 1.579, 1.093, 0.828 and 0.579 seconds, respectively. Every visible image matched the baseline dimensions and decoded RGBA hash, and all zero-area outcomes agreed.

Opening was excluded. The probe used 250-millisecond response polling, so the times include command observation and are quantized by that mechanism. Capture sizes were tested sequentially under concurrent background load. These bounded observations motivate batching in the measured pipeline. They do not isolate Office.js execution time or predict corpus throughput. A shared pixel budget can further reduce the actual number of images in a package.

H. Cost of conservative object fallback

Two retained Destatis attempts illustrate the cost of an unresolved object envelope [12]. The Elterngeld workbook's explanatory sheet accounted for 104,833 of 104,836 planned regions. At 5.29 elapsed hours, each state had completed 60,864 captures. The Abwasser workbook's contents sheet accounted for 185,856 of 185,942 regions, with 60,696 captures per state after 5.26 hours. Both attempts were still progressing. These are censored runtime observations, with no completed full-capture duration.

The planner selected the whole worksheet grid despite content envelopes of A3:A6 and A1:B47. Unsupported objects

prevented the finite-background-tail proof. In the Abwasser case, two zero-height horizontal lines had explicit stroke width. Removing them in an independently supplied A1:H50 native control changed 1,042 RGBA pixels. Geometric zero height therefore did not justify discarding their visible ink. A later bounded-line rule reduced the retained row bound to 50 in planning tests, while retaining a distant column boundary. It had not replaced either long-running attempt at the evidence checkpoint. The planning result consequently establishes no measured runtime reduction.

A separate Verkehrsunfälle diagnostic exposed 5,192,711 requested candidates before grid expansion across 208 worksheets. Combining contiguous changed row formats reduced this intermediate count to 603,267 in a fresh native repeat. All 208 content envelopes remained identical. Both diagnostics stopped after recording selection inputs, before completing rasterization. This observation supports reduced intermediate representation size, while leaving the unresolved drawing fallback and whole-workbook visual verdict open.

I. Empty context rows and a false geometry finding

A Migration development case exposed disagreement between height selection and capture selection. Worksheet `Tabelle 2.6.1` has a relevant envelope ending at row 84. The previous capture included row 85 as context. That empty row measures 12.75 points in the original and becomes hidden after cleanup. The selected `IG1:JL85` region has zero width in both states, so it yields no image. Its raw height nevertheless changes from 1,098.75 to 1,086 points, creating a geometry finding outside the relevant lower boundary.

Policy v12 clips that region to `IG1:JL84`. A fresh paired native rebuild measures 1,086 points on each side and reports no geometry or rendering difference there. The same workbook run completes 38 regions, including 37 pixel comparisons, with no capture-region differences. Separate height observations in `Tabelle 1.1` and `Tabelle 1.6` remain recorded. The control therefore resolves the identified exterior-context false finding without establishing whole-workbook equality [4].

The regression test also supplies a content envelope extending to row 96. Selection retains that lower extent, showing that the correction follows the union boundary rather than hard-coding row 84 or ignoring zero-area geometry. Historical reports keep their original policy and observations. The static collection exposes the new repeat as a separate variant. The change also demonstrates why a geometry flag and an absent image require an explanation before a finding is attributed to the transformation.

J. Capture-engine development

1) Image success without complete coverage: A synthetic million-row worksheet exercised `Range.getImage()` directly. Excel returned a PNG without an Office.js error in 2.58 seconds, but its size was only $85 \times 32{,}766$ pixels. The requested 15-point rows required 20,971,521 pixels of height. A boundary probe found 1,638 rows complete and 1,639 incomplete on that installation. This observation motivates validation of expected dimensions rather than treating successful API completion as full coverage [13].

TABLE II
SCOPE OF THE HISTORICAL NATIVE-ADAPTER EXPERIMENTS.

Adapter	Observed responsibility
VBA / GDI+	Clipboard pictures, bitmap/EMF conversion and owned-buffer pixel comparison, with optional PNG persistence
VSTO / Excel-DNA	Native drawing geometry and interoperability
Rust COM	Geometry and numeric COM property representations
Office.js	Range images, geometry and synchronization through a local host

The related original-only Lysimeter sweep covered all 978,624 requested rows of `J70953:J1048576` in 576 images, taking 814.6 seconds after readiness. Only hashes and dimensions were retained. This was a complete one-column capture workload, not an original/transformed visual comparison. Its cost illustrates why native selection and image completeness must be evaluated separately.

2) Historical method selection and failed alternatives: The earlier experiment suite remains on its own versioned branch. It exposes individual CLI experiments as well as a GUI and retains a declared 62-case protocol. Table II separates the questions answered by its adapters. They cannot all be ranked as interchangeable screenshot engines [14], [15].

In an earlier selective-overflow, 16-MP screen, five complete compatible pairs had a median Rust/.NET host workload ratio of approximately 3.34. This measures the implemented transport and telemetry paths. Six complete .NET pairs gave a median selective/heuristic overflow workload ratio of approximately 1.089. The historical decision accepted this cost for explicit intervention and restoration evidence. Neither ratio isolates a language runtime or ranks the later production pipeline.

The first controlled image-budget trial completed all 18 cases and selected 32 MP as the smallest budget within 5% of the fastest sum of per-scope median workloads. The sums were 254.45, 239.93 and 232.34 seconds for 16, 32 and 64 MP. The replication produced 245.14, 235.70 and 230.59 seconds and made the same choice. In both trials, 64 MP was fastest and had no observed budget-block failures. It was not classified as unstable. These trials used historical gridline and overflow policies, which remain part of their interpretation.

The complete replication attempted all 62 nominal cases, with 48 complete and 14 failed. The repeated failures involved four unresolved worksheet drawing-bound outcomes. They do not represent 14 independent crashes. Six of eight coverage workbooks completed, with 462 of 466 requested worksheets captured. The unresolved cases were retained instead of forcing a smaller capture or accepting missing images as equality.

3) Parallel capture pairs and memory admission: Earlier independent-capture trials on a synthetic 1,000-row workbook each completed one concurrent capture while the other add-in

exceeded its 120-second startup deadline. The repeated failure persisted after a runtime update. Those trials supplied no valid parallel throughput result. The later pair experiment used a shared inactive desktop and one local host with separate owned pair states. Its successful observations therefore concern a different coordination configuration, without isolating which change resolved the earlier startup failure [16].

A separate historical hospital selection compared one and two independent Excel pairs, each pair owning two Excel processes. With three repetitions per completed configuration, median capture-only throughput rose from 3.752 to 7.030 pairs/minute for binary PNG transport, and from 7.590 to 12.312 for hash-first transport. Peak measured job memory rose from 1,525 to 2,685 MiB and from 1,472 to 2,666 MiB, respectively [17].

Every selected tile was equal in that workload, favoring hash-first transport, which uploaded no PNG bytes. All 36 observations requested for three through eight pairs were rejected before Excel launch by RAM admission. They establish neither zero throughput nor an Excel scaling ceiling. Setup and capture were separate endpoints. One completed run spent 71.8 seconds in startup and only 7.9 seconds in capture, so capture-only rates do not predict campaign duration.

4) Adaptive geometry and hidden regions: The FRPP development workbook contains 307,920 rows, of which 304,140 are hidden. Among 19,248 original grid requests, 18,872 have zero area. A read-only COM geometry probe drove the Rust adaptive planner to 64 leaves after 710 probes in 2.815 seconds. This verifies the planner against the observed geometry. It was not a completed multi-state Office.js comparison [18].

A separate two-sheet native control exercised visible splitting and hidden regions. It retained one zero-area observation per state without inventing an image. A second control made a row visible only in the transformed state and reported rendering and geometry differences with `pixels_compared=false`. These controls support the execution and outcome contracts in the method panels, while leaving corpus-wide selector recall to independent reference tests.

K. Operational failure controls

Long-run execution has failure modes outside pixel comparison. One investigation retained 3,085 add-in registrations, and a separate checkpoint retained 617. A backed-up removal control was followed by a successful small workload. However, restoring removed registrations also produced fast startup in the later countercheck [19]. The sequence establishes an operational investigation and a counterexample to a simple accumulation explanation. It does not identify registry cleanup as the cause of a general speed improvement. Profile ownership and registration maintenance therefore retain separate mechanisms and evidence.

Historical native-adapter experiments also have bounded scope. Retained VBA adapters exercise clipboard-based picture capture and memory or persisted-image comparison. VSTO and Excel-DNA trials concern drawing geometry. The Rust-COM adapter required support for single-precision numeric shape properties in addition to double-precision and integer variants [14]. These findings explain interoperability constraints. They do not rank complete capture engines under a matched workload. No retained trace establishes the recalled clipboard-corruption and reboot sequence, so it supplies no empirical result here. Office.js batching and independent-process opening have their separately reported measured endpoints.

L. Current verification boundary

The portable-viewer smoke experiment retained a deliberately different pair, loaded its images through relative static assets and displayed its diagnostics without a running control API. An empty-partition test verified explicit selected and excluded counts without opening Excel. These tests verify execution and reporting boundaries. They do not establish selector recall.

Evaluation freeze pending. Complete the independent mutation matrix, seeded native reference sample and matched cost tables before making detection or speedup claims. Retain unresolved development cases and the original pixel observations when a later retry or software fix changes their classification.

M. Bounded comparator pilot

A development pilot supplied five frozen synthetic workbook pairs to xlCompare 12.5.0.0 and DiffEngineX 3.18.0.0 through their report commands. Both exports contained an edit at B2000 beyond the initial viewport. Neither tested export listed the hidden B3000 edit. Direct-presentation and sparse-formatting pairs produced no exported cell differences under these commands. For the dependent presentation pair, both exported the edited B1 precedent without a separate conditional-formatting or chart-rendering finding. The xlCompare trial reported a first-ten-rows export limit. These observations describe the tested reports and options, with incomplete edition and option coverage.

A subsequent DeltaSieve run used the same ten hash-verified input files and completed all five pairs under the paired in-place rebuild policy. The identical pair was excluded by byte identity. Four pairs produced native pixel differences. The offscreen pair captured two regions, detecting the B2000 edit with 47 changed pixels while the region containing hidden B3000 rendered equally. The direct presentation pair produced 7,057 changed overlapping pixels. The dependent pair produced 3,950 changed grid pixels and 36,505 changed chart pixels, including the conditional-fill and pie-chart changes. The sparse-formatting pair produced 1,281 changed pixels. Equal images were discarded with retained hash evidence.

An independent reference pass then rebuilt and rendered the same five input pairs over externally supplied rectangles, bypassing the candidate selector and content clipping. The three short pairs used Data!A1:N30, including the embedded pie chart. The offscreen and sparse pairs used continuous 256-row tiles through rows 3072 and 2304, respectively. All five reference comparisons completed. The identical control

remained pixel-equal. Only the tile containing B2000 changed in the offscreen pair, while the tile containing hidden B3000 remained equal. Only the tile containing J2099 changed in the sparse pair. The direct and dependent pairs also differed in the broader reference rectangles.

The reference observations agree with the selected run at pair level. They do not estimate corpus recall or pixel-union coverage. Reference coverage remains limited to the supplied rectangles. Pixel totals from different capture extents or standalone and embedded chart rendering are not directly interchangeable. No matched comparative speedup is established by these single runs under concurrent background load. Source identities and mutation classes are shared with the bounded comparator pilot, while the measurement protocols and reported evidence remain distinct.

Two separately prepared large Destatis pairs contained 107 and 215 worksheets. The observed xlCompare wall times were 53.748 and 21.711 seconds, respectively. DiffEngineX required 131.692 and 161.221 seconds. Its reports represented every source worksheet and reported zero cell differences in formula-comparison mode. These were single trials under concurrent development load. Companion Excel process resources were not measured over their full lifetime, and no matched DeltaSieve cost trial was completed. These measurements therefore establish bounded execution feasibility and report coverage, leaving comparative cost and rendered-equivalence claims open.

N. Check Performance preparation checkpoint

A completed Destatis preparation campaign attempted all 1,182 source inputs and retained 1,016 prepared outputs and 166 failures. Its raw failure reasons contain 165 opening timeouts and one automation failure when setting Excel's Interactive property. These are preparation outcomes. Visual validation of the outputs and matched baseline cost measurements remain open.

Recovery checks of the same 166 frozen failure inputs used an owned opening-dialog monitor. The first three inputs completed in 13.213 seconds and the remaining 163 in 442.761 seconds, each batch with two workers. The checks identified Name Conflict dialogs for all 165 former opening timeouts. The Interactive-property failure recurred. A separate known-positive input produced its prepared output in 8.114 seconds. Source-identity reconciliation accounts for every original failure, while conflicting-name contents remain unresolved. Worker count changed during the first campaign, so its aggregate elapsed time is observational. Original and recovery receipts remain separately versioned alongside the 1,016 successful preparation artifacts.

O. Bounded graphical-object controls

Embedded objects expose a distinct coverage boundary. Excel reports the retained Word preview and PDF icon as Unsupported shapes, while range images render their cached previews. Two explicit PDF-icon mutation controls used the same fixed 883-by-429-pixel reference area. Moving the frame changed 6,473 RGBA pixels; replacing only its cached EMF preview changed 5,242. The captures completed in 10.769 and 10.395 seconds, respectively. Package-member checks confirmed exactly the intended mutations, and the rendered images were inspected.

A separate retained workbook contains three rectangular Text 1 shapes with identical from/to anchors, zero-size transforms and clipped text overflow. Removing those three anchors yielded identical full-area images on Deutschland, West and Ost VZ. Enlarging the same shapes yielded 1,531, 1,737 and 1,487 changed pixels, respectively, with identical dimensions within each pair. The removal and expansion batches completed in 12.728 and 12.862 seconds. Their independently supplied areas and relationship-resolved sheet/drawing mapping remain frozen. An initial ordinal-based fixture mapping was corrected and retained as an incomplete setup attempt.

These development controls establish renderer sensitivity for the stated mutations and areas. A bounded OLE coverage rule still requires supported namespace and property checks, consistent package identities, anchor agreement, native-frame agreement and complete object accounting. The current reference guard retains the unsupported outcomes explicitly. These observations contribute to the declared mutation matrix; corpus-level detection coverage remains an independent evaluation obligation.

P. Repeated column widths in a uniform background

The retained Destatis workbook `anbau-gartenbaugewaechse-2030224169005.xlsx` exposed excessive background coverage. Its rebuilt original contains 128,912 column definitions, compared with 2,197 in the rebuilt SheetShrink output. In 166 worksheets, the column styles apply one uniform background across the full worksheet width. For example, sheet `1901.01` contains serialized cells only through `J166`, but the former plan also selected distant columns such as `ACO`. Of 16,101 measured regions across 164 selected sheets, 15,936 lay wholly outside the serialized-cell envelopes. That count alone is diagnostic, since empty cells can still have visible formatting effects.

The uniform-background guard yields 164 coalesced XML-selected regions for the retained rebuilt partners. A fresh native comparison produces 165 measured regions, with identical images for all 165 pairs. Capture selection and transport took 41.371 seconds and image comparison took 1.379 seconds in this single trial. The earlier large-plan attempt remained active for more than 40 minutes before interruption. This is a development observation under concurrent load, rather than a matched end-to-end speedup experiment. Neither identical selected images nor the reduced plan establishes exhaustive workbook equivalence. Regression controls retain distant fill transitions, borders, unknown styles, explicit cell overrides and unsupported conditional-formatting contexts.

A second retained Enron input, `5d98c82491876616a27c81387d370975402de30439051002d64c0c1f6e5f3d68.xlsx`, contains one worksheet named `Feb`. Its rebuilt original has 545,958 serialized cells

through row 49,633, compared with 5,341 cells through row 486 in the rebuilt transformed state. Both retain an empty AutoFilter over A3:O485. The former unsupported-filter fallback expanded background coverage to the full worksheet width and produced 65,664 measured regions per state. At an observed checkpoint after about 50 minutes, approximately ten percent of those regions had been captured. The attempt was interrupted with its evidence retained. The bounded-empty-filter guard reduces the XML-selected union to A385:P49634, represented by 385 internal grid tiles before native image-size partitioning. The full serialized-cell envelope remains in scope. A fresh native run yields 73 measured regions per state, all with identical images. Capture selection and transport take 28.114 seconds and image comparison takes 0.212 seconds. The native partition depends on measured geometry and is therefore distinct from the 385 XML grid tiles. These single development runs retain the full serialized-cell envelope and do not establish corpus recall or a matched end-to-end speedup.

Q. Distinguish geometry work from an idle worker

The Destatis workbook gestorbene-alters-und-geburtsjahre-2010193039005.xlsx exposed a watchdog accounting defect. The retained interrupted attempt was receiving successful native height replies, including one for sheet AJ 1955 O-1, but the capture-progress counter did not include those replies. The 120-second idle watchdog consequently interrupted active measurement work before image capture. The correction advances progress after height-run validation. This changes failure detection and preserves the selected visual evidence contract. The native repeat completed in 408.453 seconds with 741 identical image pairs. Capture selection and transport took 219.786 seconds, and image comparison took 8.863 seconds. The original interrupted observation and successful repeat remain separate records. This case demonstrates active measurement beyond the former idle deadline without extending it on heartbeat-only activity.

R. Repeat a small native chart discrepancy

One Enron workbook, 00f8fb769c305d91 14647758f3c38cfa2cbbcc18dc5bb9ad 7b254b2b5949506b.xlsx, initially produced 26 changed RGB pixels with maximum channel difference 36 in the embedded Chart 1 on Hedge Example. The corresponding worksheet image was identical. A package inspection found no worksheet formulas in this workbook. A separate native control compared two inputs with identical uncompressed package members and a different ZIP comment, avoiding the byte-identity shortcut. Conservative chart capture produced identical images for both the embedded chart and chart sheet Chart3; that control took 20.604 seconds. A fresh original-versus-SheetShrink comparison also produced identical worksheet and chart images. The earlier discrepancy remains retained as an unreproduced native rendering observation. These controls justify neither a global pixel tolerance nor exclusion of another chart's differences.

Additional same-content controls for Enron inputs 106bb183b73704d6... and 806f1a068f4e9ca4... reproduced two changed pixels in Run Query, Diagramm 4, and seven in Results, Chart 1, respectively. Maximum channel differences were 35 and 36 out of 255. Each control changes only a ZIP comment between its inputs and verifies SHA-256 equality of every uncompressed member. Conservative chart mode forces native chart exports. After rebuilding, worksheet, style and chart parts remain byte-identical across the two states, and chart geometry is equal. Remaining package differences concern workbook metadata, core properties and, in the first case, the calculation chain. These controls demonstrate native path variability without applying SheetShrink. They do not establish that every small chart discrepancy has the same cause or justify a pixel tolerance.

S. Controlled opening recovery

Six selected Destatis workbooks displayed conflicts between legacy print names and canonical built-in definitions. Controlled copies preserve both definitions under distinct names after checking for alias references. Five comparisons completed in the initial recovery batch. Four had identical captured regions. The workbook pruefungen-hochschulen-2110420217005.xlsx retained one pixel discrepancy on TAB-04, requiring subsequent investigation. The sixth workbook, bevoelkerungsbewegung-2010110157005.xlsx, exposed a conservative drawing fallback on sheet 2.11. Its stored line top was approximately 960.5 points, while Excel reported approximately 917.529 points. Both observations fit within the measured A1:K64 rectangle, with the required stroke margin. These recovered-baseline observations retain separate input identities.

Five selected Enron workbooks also displayed native repair prompts when tested with their original input as the sole opened comparison input. Their SHA-256 identities and prompt evidence are retained. Such incomplete opening attempts provide no image-equivalence verdict.

The repeated TAB-04 discrepancy affected 25,640 pixels in a 784-by-11,727-pixel image. A same-original reference control reproduced the original image hash exactly. Repeating original versus SheetShrink reproduced both earlier image hashes. Restoring 87 removed empty cells carrying the explicit default font and no-fill metadata made the full reference image pixel-identical. After extending font-representative eligibility for this metadata combination, a newly generated SheetShrink output also reproduced the original image hash. The independent reference plan was TAB-04!A1:L916. The successful fixed-output reference check took 12.798 seconds. Failed intermediate counterfactual packages were rejected during XML validation and were excluded from these observations.

A separate Binnenschifffahrt case, statistischer-bericht-binnenschifffahrt-

2080400231025.xlsx, completed in 143.709 seconds after setup-phase progress was included in watchdog accounting. Its original opening/saving control recorded 4,422,761 XML change entries, compared with 3,088 input transformation changes. The direct rebuilt comparison selected 13 image pairs, all identical. Selection and transport took 3.004 seconds and image comparison took 0.235 seconds.

The remaining Bevoelkerungsbewegung selector fallback involved two legacy textboxes on 3.13 with zero native width and height. Their XML falls outside the closed line recognizer. An independently frozen reference plan therefore rendered 622 finite areas across all 96 worksheets in 227.099 seconds. The plan used the union of the serialized input/output dimensions, four context rows and two context columns, partitioned independently into at most 16 columns by 128 rows. In 529 areas every common RGBA pixel was identical and any unpaired pixels exactly repeated the shared boundary row or column. In the other 93 areas, one state had zero renderable height while the retained counterpart was entirely opaque white. No changed overlapping pixels or nonrepeating retained edge content remained. These observations cover the stated finite plan. The unsupported automatic textbox qualification and its rejected full-grid plan remain recorded separately.

A controlled Enron opening experiment isolated a metadata inconsistency in one converted workbook. Renumbering duplicate sheet identifiers alone still failed, as did removal of its calculation chain alone. A copy combining unique sheet identifiers with removal of the optional calculation chain opened in 0.841 seconds. The original XLS input itself opened in 0.693 seconds, but its freshly saved XLSX also reproduced the reopening failure. This establishes a conversion baseline issue for that case. Controlled recovery copies preserve worksheet, style, shared-string and drawing parts byte-for-byte and require native rebuild before visual comparison. Their identifiers and changed package members are retained explicitly.

T. Observe post-capture comparison progress

The Data.gov workbook 38hospitaldata_5d43543005e1.xlsx reached all 3,546 captures in both rebuilt lanes before an earlier attempt exceeded its 120-second idle deadline during image comparison. The full interrupted attempt lasted 1,645.739 seconds. Its original rebuilt lane recorded 1,349.828 seconds in capture phases. The comparison phase lacked per-image work notifications. The correction marks that phase active and advances verified progress after each completed image-pair verdict. The interrupted attempt has no completed visual verdict. The failure-retention change was separately exercised through the normal runner on a known native opening failure. Its failed record retained a working-evidence directory and the native blocking-dialog receipt, while the existing partial-report test confirmed that incomplete observations cannot be reused as completed comparisons.

The repeat with verified per-image comparison progress completed all 3,546 selected image pairs without a pixel difference. It lasted 1,767.483 seconds, including 1,335.191 seconds for capture selection and transport and 245.093 seconds for image comparison. The latter exceeds the earlier 120-second idle deadline even though comparison work was advancing. This repeat establishes successful completion of the selected plan and quantifies the remaining image processing cost. It does not claim that the entire capture plan is minimal. The rebuilt input also has a genuinely broad serialized value footprint. Cost Allocation Data contains 402,582 cells with value elements through column 4,315, and Financial and Utilization Data contains 1,263,626 through column 12,816. Both extend through row 452 and contain no cell formulas. These counts describe serialized values, including any stored empty strings. They distinguish this wide-table cost case from a plan expanded solely into an empty full-grid background. Safe further selection reduction remains open.

The current study cohort freezes the 49 whole workbooks with greatest absolute compressed-file byte reduction in each of Destatis, Data.gov and Enron. This is a stress selection of 147 workbook identities. The cohort uses workbook-level ranking. Individual worksheet reductions do not select additional inputs. Every processor uses the same selected original identities. Earlier observations from the larger development selection remain provenance. A missing processor output is recorded as unavailable, separately from a failed visual comparison.

U. Finite conditional formatting within a repeating background

In kulturfinanzbericht-gesamtmaterial-1023003229005.xlsx, worksheet Tabelle 4.8-1 contains four finite color-scale application areas and borderless inherited solid fills. Requiring every rule to be inactive on blank cells previously expanded the potential background extent to the full grid. An XML-only retained-input check of the revised enclosure produced A1:P79, while retaining all four conditional-format blocks and without classifying their blank cells as inactive. Focused regression cases distinguish an application inside the content envelope, one extending beyond the content, and a full-grid application. The XML enclosure is a planning observation. Its native comparison outcome is reported separately when available.

V. Inspect retained chart-pixel observations

Figure 8 illustrates a retained development finding from the Data.gov Leachate workbook, worksheet Fig. S2, Chart4. The report records 20 changed pixels, a maximum channel change of 47 out of 255 and an absolute channel-difference sum of 1,251. The small affected regions are visible in the magnified diagnostics. Their size alone does not identify their cause. Rendering variation remains a hypothesis for this example until supported by independent repeated native controls. The finding and its image evidence remain available in the report. This historical illustration is separate from final 49-workbook cohort counts.

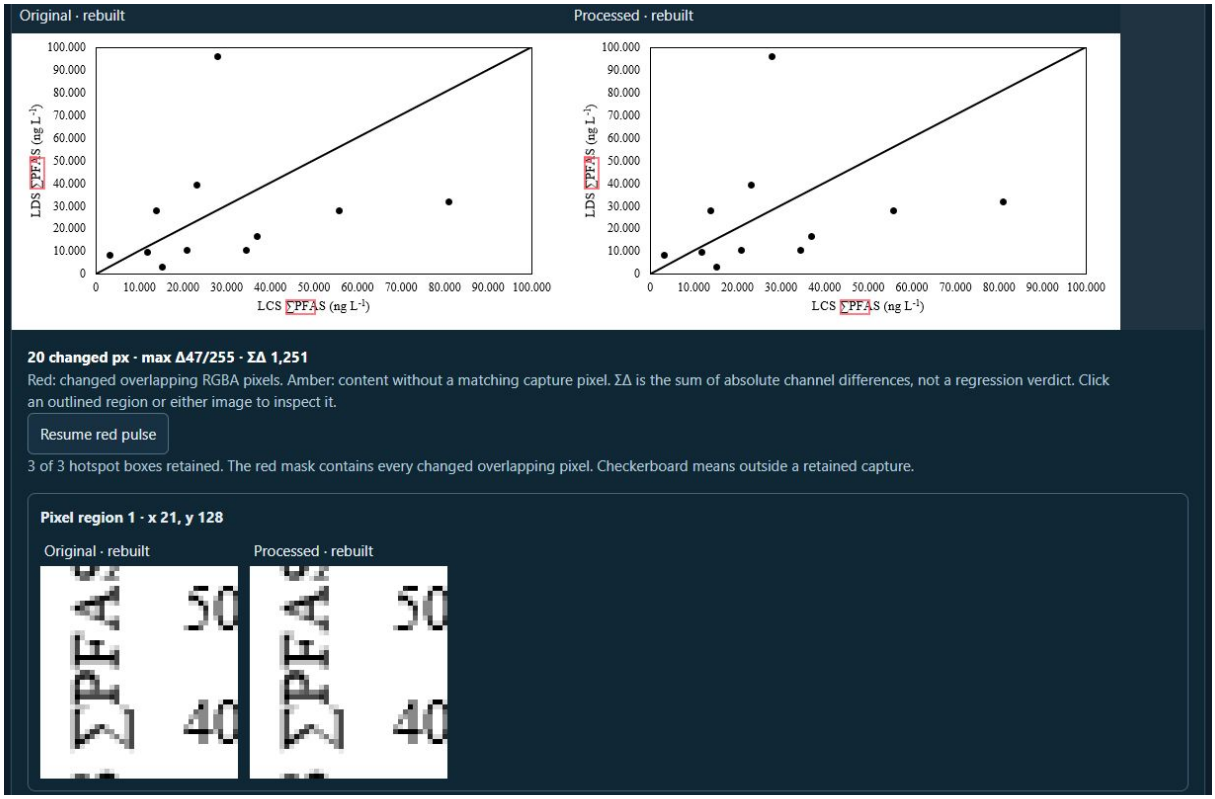


Fig. 8. Retained native chart comparison with 20 changed pixels. The two chart images, pixel metrics and magnified diagnostics allow inspection of a small observed discrepancy. A small discrepancy is insufficient evidence to attribute the change to rendering or to exclude it from the report. The screenshot is cropped for legibility; the original image evidence is retained.

W. Separate save-control volume from capture count

The Data.gov workbook `MORB_Data_b2b3bf4db76f.xlsx` initially exceeded a 120-second idle deadline during pre-capture XML inspection. Its provisional 237,733 candidates were not the final native image plan. A bounded single-input repeat with a 600-second idle deadline completed in about 1,032 seconds. It retained 514 worksheet regions and one chart per rebuilt state, with no reported image difference. Capture selection and transport took 246.189 seconds, and image comparison took 33.852 seconds.

The two ordinary-save controls contained 14,274,504 and 14,274,497 serialized XML changes. Their measured audit stages took 84.984 and 70.627 seconds, respectively. These control counts explain substantial non-rendering work; they must not be presented as requested captures. A subsequent guard omits legacy coordinate projection when the paired execution has no pre-rebuild original image state. A focused no-Excel regression exercises that absent-state case. The retained native repeat predates the guard and establishes neither a measured speedup from it nor exhaustive worksheet coverage.

X. Unfilled columns beneath a repeating row band

The Kulturfinanzbericht worksheet `Tabelle 8.2-1` produced 127 selected regions under policy version 14. Of these, 125 far-tail findings depicted a horizontal grey row band with different capture widths. Column records changed widths and font styles, but introduced no visible column fill or border. The revised XML rule bounds the retained rebuilt states to `A1:BO114` and `A1:BO61`. Every explicit cell remains included. Focused cases preserve distant explicit cells and finite fill transitions and reject border or unknown-style qualification.

An independent native `A1:BP114` comparison covers the finite union and context. Both images have 4,625 by 2,080 pixels, identical SHA-256 identities and equal measured geometry. This reference completed in 14.585 seconds. The subsequent complete version-15 workbook comparison completed 350 selected regions without a pixel, geometry or rendering difference. This worksheet contributes one region, compared with 127 previously. Whole-workbook capture selection and transport took 119.009 seconds, compared with 624.755 seconds in the retained version-14 run. Producer versions and concurrent workloads differ, so these observations are not a matched performance experiment. The original 125 far-tail observations and their images remain retained.

V. Related Work

A. Spreadsheet change inference

SheetDiff infers changes between spreadsheet versions and presents them in a single contextual view [20]. This establishes spreadsheet change inference as prior work. RowColAlign extends longest-common-subsequence reasoning to

two-dimensional alignment and evaluates edit recovery using generated pairs with planted changes [21]. Its known edit history provides a methodological precedent for controlled mutations. Our evaluation additionally requires native observations of which mutations produce visible effects under the chosen state-preparation policy. Edit recovery and rendered-effect coverage have different observation units.

B. Spreadsheet dependencies and testing

Spreadsheet slicing connects dependency analysis with integrated testing and fault localization [1]. DeltaSieve uses supported dependencies to select native observation regions. The two methods have different adequacy obligations. Formula-dependency coverage alone does not cover every visual effect.

C. Workbook comparison and native evidence

Spreadsheet Compare documents workbook comparison reports with formula and cell-format categories, including hidden sheets [2]. This is an important existing capability. DeltaSieve's comparison must examine native rendered effects, candidate selection and corpus execution explicitly, using common test inputs where the installed tools support them.

xlCompare documents command-line comparison and exported difference reports [22]. DiffEngineX documents whole-workbook and folder comparisons, report workbooks, and comparisons of formulas and constants. Its help explicitly excludes chart comparison [23]. These documented capabilities motivate a shared input pilot with exported reports. A report of edited cell inputs and an observation of their dependent rendered consequences answer different questions. The pilot must therefore record the tested options and report coverage instead of treating every comparator as an interchangeable visual oracle.

Synkronizer's edition documentation lists content, formatting and Excel-name comparison features [24]. Spreadsheets-Diff advertises two- and three-way comparison, key-column matching, and workbook/VBA comparison features [25]. These vendor descriptions identify further comparison candidates. A native pixel-observation contract and its tested coverage require separate verification for each product and version.

D. Cross-system testing and parsing

SQuaLity examines barriers to reusing database test suites across systems, including configuration and execution dependencies [26]. This suggests recording the renderer and environment when reusing observations. The analogy does not transfer database test adequacy guarantees to spreadsheet rendering. SheetReader's specialized parsing work [27] is relevant to efficient package inspection, whose cost must be separated from native rendering cost.

E. Selective regression testing and visual models

Ekstazi selects regression tests through recorded file dependencies [28]. It tracks dynamically used files and their checksums. Its separation of analysis, execution and dependency collection motivates measuring selection overhead together with avoided work. DeltaSieve's spreadsheet dependency model and native image observations require their own coverage evaluation.

ReDeCheck compares responsive web layout models and reports diagnostic screenshots [29]. It samples viewport widths and refines layout transitions. DeltaSieve instead follows workbook-state changes and supported spreadsheet dependencies. Both settings motivate distinguishing a model-level warning, an observed visual effect and a confirmed defect. Their models and validation subjects require separate experimental oracles.

VI. Discussion and Limitations

The central risk is an incomplete candidate model. A proof for supported inputs must state its dependencies and applicable namespace or rule subset. Unknown features need a capture fallback or an explicitly incomplete outcome. Runtime limits and failures reduce achieved coverage and must remain in the report.

A native renderer is also an experimental dependency. Recalculation, volatile formulas, file-location functions and rendering variability can change output without a transformation defect. A volatile dependency trace explains a supported path rather than every image difference. A white exterior canvas difference requires geometry and containment checks before classification. No global pixel tolerance is inferred from one rare anti-aliasing case.

The Observatory reduces the effort of investigating these outcomes. Portable reports and a future recorded-event replay can demonstrate execution and make findings inspectable. They do not establish authenticity solely by being detailed, and replay cannot supply missing historical telemetry. Scientific claims rely on source identities, implementation versions and the independent evaluation.

VII. Conclusion

DeltaSieve organizes targeted native spreadsheet comparison around inspectable change evidence and explicit coverage. The remaining empirical obligation is to measure detection against independent native references and to quantify cost on matched inputs. The artifact retains observations and uncertainty so those claims can be checked at a fixed reporting cutoff.

References

[1] J. Reichwein, G. Rothermel, and M. Burnett, "Slicing spreadsheets: An integrated methodology for spreadsheet testing and debugging," in *Proceedings of the Second Conference on Domain-Specific Languages*. USENIX, 1999, pp. 25–38. [Online]. Available: https://www.usenix.org/legacy/publications/library/proceedings/dsl99/full_papers/reichwein/reichwein_html/index.html

[2] Microsoft, "Basic tasks in spreadsheet compare," Microsoft Support, accessed November 8, 2026. [Online]. Available: https://support.microsoft.com/en-gb/excel/basic-tasks-in-spreadsheet-compare

[3] "SheetShrink: Efficient spreadsheet cleanup," Companion working manuscript prepared for ICDE 2027, 2026, supplementary material, related-manuscripts/sheetshrink.pdf.

[4] DeltaSieve project, “Targeted visual validation behavior, empty trailing row exclusion,” Versioned protocol and evidence, revision 7ca7c2a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/7ca7c2a/docs/behavior/targeted-validation.md

[5] ——, “Resident capture events and timings,” Versioned execution protocol, revision a7eef1a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/a7eef1a/docs/visual-validation/resident-events.md

[6] ——, “Observatory live stream experiment,” Versioned protocol and evidence, revision 7ca7c2a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/7ca7c2a/docs/visual-validation/experiments/observatory-live-stream-20261109.md

[7] ——, “Static sheet atlas,” Versioned protocol and evidence, revision 7ca7c2a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/7ca7c2a/docs/behavior/static-atlas.md

[8] Microsoft, “Excel recalculation,” accessed November 9, 2026. [Online]. Available: https://learn.microsoft.com/en-us/office/client-developer/excel/excel-recalculation

[9] NIST/SEMATECH, “Confidence intervals for a binomial proportion,” e-Handbook of Statistical Methods, Section 7.2.4.1, accessed November 9, 2026. [Online]. Available: https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm

[10] DeltaSieve project, “Native Excel startup across private desktops,” Protocol and retained findings-four.json, revision a7eef1a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/tree/a7eef1a/docs/visual-validation/experiments/parallel-workbook-open

[11] ——, “Batched Office.js geometry and image captures,” Protocol and results, revision a7eef1a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/tree/a7eef1a/docs/visual-validation/experiments/batched-capture-geometry

[12] ——, “Native legacy-object bounds in large destatis plans,” Investigation and retained controls, revision a7eef1a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/a7eef1a/docs/visual-validation/investigations/native-legacy-object-bounds-20261109.md

[13] ——, “Office.js image height and tall-range hash sweep,” Versioned development evidence, revision 1b9ee98, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/1b9ee98/docs/visual-validation/experiments/officejs-image-limit/README.md

[14] ——, “Visual validation experiments and native adapters,” Versioned protocol and evidence, revision ce0dc7b, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/ce0dc7b/docs/visual-validation/experiments/README.md

[15] ——, “Capture method decision and protocol replication,” Versioned development evidence, revision ce0dc7b, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/ce0dc7b/docs/visual-validation/decisions.md

[16] ——, “Concurrent capture startup failures and paired follow-up,” Versioned development evidence, revision ce0dc7b, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/ce0dc7b/docs/visual-validation/experiments/concurrent-office-capture/README.md

[17] ——, “Paired comparison throughput and admission observations,” Versioned development evidence, revision ce0dc7b, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/ce0dc7b/docs/visual-validation/experiments/concurrent-office-capture/observations.md

[18] ——, “Adaptive capture partition and native controls,” Versioned development evidence, revision 1b9ee98, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/1b9ee98/docs/visual-validation/experiments/adaptive-capture/README.md

[19] ——, “Add-in startup investigation and reversible controls,” Versioned protocol and evidence, revision 7ca7c2a, 2026. [Online]. Available: https://github.com/Alexander-Aue-Johr/excel-shrink-rust/blob/7ca7c2a/docs/visual-validation/investigations/addin-startup-20261108.md

[20] C. Chambers, M. Erwig, and M. Luckey, “SheetDiff: A tool for identifying changes in spreadsheets,” in *IEEE Symposium on Visual Languages and Human-Centric Computing*, 2010, pp. 85–92. [Online]. Available: https://web.engr.oregonstate.edu/~erwig/papers/SheetDiff_VLHCC10.pdf

[21] A. Harutyunyan, G. Borradaile, C. Chambers, and C. Scaffidi, “Planted-model evaluation of algorithms for identifying differences between spreadsheets,” in *IEEE Symposium on Visual Languages and Human-Centric Computing*, 2012, pp. 7–14.

[22] Spreadsheet Tools, “xlCompare command line parameters,” accessed November 9, 2026. [Online]. Available: https://xlcompare.com/excel-compare-command-line-parameters.html

[23] Florencesoft, “Detailed DiffEngineX help,” accessed November 9, 2026. [Online]. Available: https://www.florencesoft.com/xldiffhelp.html

[24] Synkronizer, “Synkronizer 11 editions,” vendor documentation, accessed November 9, 2026. [Online]. Available: https://help11.synkronizer.com/intro_editions.htm

[25] SpreadsheetsDiff, “Compare and merge Excel workbooks,” vendor description, accessed November 9, 2026. [Online]. Available: https://spreadsheetsdiff.com/

[26] S. Zhong and M. Rigger, “Understanding and reusing test suites across database systems,” *Proceedings of the ACM on Management of Data*, vol. 2, no. 6, 2024.

[27] H. Gavriilidis, F. Henze, E. T. Zacharatou, and V. Markl, “SheetReader: Efficient specialized spreadsheet parsing,” *Information Systems*, 2023.

[28] M. Gligoric, L. Eloussi, and D. Marinov, “Practical regression test selection with dynamic file dependencies,” in *Proceedings of the International Symposium on Software Testing and Analysis*, 2015, pp. 211–222. [Online]. Available: https://users.ece.utexas.edu/~gligoric/papers/GligoricETAL15Ekstazi.pdf

[29] T. A. Walsh, G. M. Kapfhammer, and P. McMinn, “ReDeCheck: An automatic layout failure checking tool for responsively designed web pages,” in *Proceedings of the International Symposium on Software Testing and Analysis*, 2017, pp. 360–363.